

SEC5L: A Five-Field Mixed Tetrahedral Finite Element on the Symmetric Elasticity Complex with Logarithmic Strain

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September 24, 2026

Abstract

We propose the SEC5L element, a tetrahedral finite element built on the symmetric elasticity complex (SEC) with 5 independent discrete fields and a logarithmic strain measure. The five fields are displacement, volumetric log-strain, deviatoric log-strain, hydrostatic pressure, and deviatoric Kirchhoff stress. The formulation is structured so that all finite-deformation nonlinearity is absorbed into the kinematic relation $\boldsymbol{\epsilon} = \frac{1}{2} \log(\mathbf{F}^T \mathbf{F})$, while the constitutive relation is algebraically linear in log-strain space. The orthogonal volumetric–deviatoric decomposition $\boldsymbol{\epsilon} = \frac{1}{3} \theta \mathbf{I} + \mathbf{e}$, with $\theta := \text{tr } \boldsymbol{\epsilon}$ and $\mathbf{e} := \text{dev } \boldsymbol{\epsilon}$, and $\boldsymbol{\tau} = p \mathbf{I} + \mathbf{s}$, with $p := \frac{1}{3} \text{tr } \boldsymbol{\tau}$ and $\mathbf{s} := \text{dev } \boldsymbol{\tau}$, is native to the Hencky measure, which decouples the J_2 plastic return map (pointwise on $\mathbf{e}$) from the volumetric response and exposes a natural handle for stiff bulk-wave regimes. The independent strain and stress fields remove the cross-space constitutive evaluation that compromises stabilized displacement formulations, and all four auxiliary fields admit exact L^2 projections from $\boldsymbol{\varphi}$ with mass matrices that are constant in the reference configuration and pre-factorizable, reducing the global nonlinear solve to a displacement-sized problem. The formulation is intended as an alternative to the composite tetrahedron of Ostien et al. [15] and its five-field reformulation [9], targeting near-incompressible finite-deformation elasticity, isochoric plasticity, and weak-shock regimes with large bulk-to-shear ratio.

Keywords: SEC5L, Hu-Washizu, tetrahedron, mixed formulation, logarithmic strain, Kirchhoff stress, symmetric elasticity complex, plasticity.

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1 Introduction

Tetrahedral finite elements are the natural choice for discretizing solid bodies with complex geometry because they admit robust automatic meshing. Their performance in problems involving near-incompressible materials, isochoric plasticity, and stiff bulk-wave regimes has, however, long been the limiting factor in their adoption for large-deformation Lagrangian solid mechanics [8, 10, 16, 17, 19]. Stabilized displacement methods [11, 12, 17] and assumed-gradient schemes such as the composite tetrahedron of Ostien et al. [15] have made significant progress. The composite tetrahedron with mean-dilatation patch targets isochoric plasticity and near-incompressible response through a three-field Hu-Washizu functional with element-local discontinuous strain and stress and static condensation. The subsequent five-field reformulation of Foulk III et al. [9] adds independent Jacobian and pressure fields to address a deleterious soft mode that arises in constant-pressure formulations.

The present formulation departs from the composite-tetrahedron lineage in three specific ways. First, the independent strain and stress fields are discretized in *conforming* symmetric-tensor spaces compatible with the symmetric elasticity complex [2, 5], giving pointwise Cauchy symmetry of the discrete stress and normal-traction continuity across element faces as structural properties of the finite element spaces rather than of the constitutive law. Second, the finite-deformation measure is the Hencky log-strain $\boldsymbol{\epsilon} := \frac{1}{2} \log(\mathbf{F}^T \mathbf{F})$, with its work-conjugate Kirchhoff stress $\boldsymbol{\tau}$; the Hencky decomposition $\boldsymbol{\epsilon} = \frac{1}{3}(\log J)\mathbf{I} + \mathbf{e}$ is orthogonal in the Frobenius inner product and separates naturally into independent volumetric and deviatoric discrete fields. Third, the constitutive law is *algebraically linear* in $(\boldsymbol{\epsilon}, \boldsymbol{\tau})$ for isotropic hyperelastic response, so the radial return map for J_2 plasticity reduces exactly to the small-strain algorithm of Simo [18] acting pointwise on the deviatoric strain at quadrature.

The combination of these choices yields a formulation in which the four auxiliary fields are obtained from the displacement field by L^2 projections with mass matrices that are constant in the reference configuration. Pre-factorization at setup reduces each Newton iteration to a displacement-sized nonlinear solve, while the auxiliary fields preserve the spatial-discretization correctness properties (symmetry, traction continuity) through their conforming finite element spaces. Dynamic and quasi-static regimes use the same spatial formulation; the mass matrix enters only through the displacement block.

2 Continuum Formulation

Consider a body $B \subset \mathbb{R}^3$ undergoing a motion described by $\mathbf{x} = \boldsymbol{\varphi}(\mathbf{X}, t) : B \times [t_1, t_2] \rightarrow \mathbb{R}^3$, with deformation gradient $\mathbf{F} := \text{Grad } \boldsymbol{\varphi}$ and Jacobian $J := \det \mathbf{F}$. The boundary ∂B with outward unit normal $\mathbf{N}$ is partitioned into a displacement boundary $\partial_\varphi B$ and a traction boundary $\partial_T B$. Prescribed boundary data are $\boldsymbol{\chi} : \partial_\varphi B \times [t_1, t_2] \rightarrow \mathbb{R}^3$ and $\mathbf{T} : \partial_T B \times [t_1, t_2] \rightarrow \mathbb{R}^3$. Let $R : B \rightarrow \mathbb{R}$ be the reference mass density and $\mathbf{B} : B \times [t_1, t_2] \rightarrow \mathbb{R}^3$ the body force per unit mass.

2.1 Hencky Log-Strain and Kirchhoff Stress

Let $\mathbf{C} := \mathbf{F}^T \mathbf{F}$ be the right Cauchy–Green tensor. The material Hencky log-strain is

$$\boldsymbol{\epsilon} := \frac{1}{2} \log \mathbf{C} \in \mathbb{S}, \quad (1)$$

evaluated numerically by an inverse scaling-and-squaring Padé approximant applied to $\mathbf{C}$. Its orthogonal volumetric–deviatoric decomposition is

$$\boldsymbol{\epsilon} = \frac{1}{3} \theta \mathbf{I} + \mathbf{e}, \quad \theta := \text{tr } \boldsymbol{\epsilon} = \log J, \quad \mathbf{e} := \text{dev } \boldsymbol{\epsilon}. \quad (2)$$

The work-conjugate stress measure is the Kirchhoff stress $\boldsymbol{\tau}$, symmetric and defined such that $\boldsymbol{\tau} : \dot{\boldsymbol{\epsilon}}$ equals the stress power per unit reference volume for isotropic material response. Its decomposition mirrors (2):

$$\boldsymbol{\tau} = p \mathbf{I} + \mathbf{s}, \quad p := \frac{1}{3} \text{tr } \boldsymbol{\tau}, \quad \mathbf{s} := \text{dev } \boldsymbol{\tau}. \quad (3)$$

2.2 Stored Energy and Constitutive Law

For isotropic hyperelasticity, the Helmholtz free-energy density is additively decomposable in the Hencky measure:

$$A(\boldsymbol{\epsilon}) = W^{\text{vol}}(\theta) + W^{\text{dev}}(\mathbf{e}), \quad W^{\text{vol}}(\theta) = \frac{\kappa}{2} \theta^2, \quad W^{\text{dev}}(\mathbf{e}) = \mu \|\mathbf{e}\|^2, \quad (4)$$

with μ the shear modulus and κ the bulk modulus. The constitutive law is *algebraically linear*:

$$p = \frac{\partial W^{\text{vol}}}{\partial \theta} = \kappa \theta, \quad \mathbf{s} = \frac{\partial W^{\text{dev}}}{\partial \mathbf{e}} = 2\mu \mathbf{e}. \quad (5)$$

Extensions to elasto-plasticity replace (5)₂ by

$$\mathbf{s} = 2\mu (\mathbf{e} - \mathbf{e}^{\text{p}}), \quad (6)$$

with $\mathbf{e}^{\text{p}}$ the deviatoric plastic log-strain evolved by a J_2 radial return map described in §5.

2.3 Five-Field Hu-Washizu Functional

Introduce independent strain fields $\bar{\theta}$ and $\bar{\mathbf{e}}$, and enforce their equality with the pointwise kinematic fields $\theta(\boldsymbol{\varphi})$ and $\mathbf{e}(\boldsymbol{\varphi})$ weakly through Lagrange multipliers $\bar{p}$ and $\bar{\mathbf{s}}$. At a stationary point the multipliers coincide with the work-conjugate stress fields: the hydrostatic pressure and the deviatoric Kirchhoff stress, respectively. The Hu-Washizu functional is

$$\begin{aligned} \Phi[\boldsymbol{\varphi}, \bar{\theta}, \bar{\mathbf{e}}, \bar{p}, \bar{\mathbf{s}}] &:= \int_B W^{\text{vol}}(\bar{\theta}) \, dV + \int_B W^{\text{dev}}(\bar{\mathbf{e}}) \, dV \\ &+ \int_B \bar{p} (\theta - \bar{\theta}) \, dV + \int_B \bar{\mathbf{s}} : (\mathbf{e} - \bar{\mathbf{e}}) \, dV \\ &- \int_B R \mathbf{B} \cdot \boldsymbol{\varphi} \, dV - \int_{\partial_T B} \mathbf{T} \cdot \boldsymbol{\varphi} \, dS, \end{aligned} \quad (7)$$

with θ and $\mathbf{e}$ understood as pointwise functions of $\boldsymbol{\varphi}$ through (1)–(2).

2.4 Variations and Euler-Lagrange Equations

Assume $\boldsymbol{\varphi} \in U := (W_2^1(B))^3$, $\bar{\theta}, \bar{p} \in Q := L^2(B)$, $\bar{\mathbf{e}} \in V_e := H(\text{Curl}; B; \mathbb{S}^{\text{dev}})$, and $\bar{\mathbf{s}} \in V_s := H(\text{Div}; B; \mathbb{S}^{\text{dev}})$, where $\mathbb{S}^{\text{dev}}$ denotes the space of symmetric trace-free second-order tensors. Test functions are $\boldsymbol{\psi} \in U$ with $\boldsymbol{\psi} = \mathbf{0}$ on $\partial_\varphi B$, $\vartheta, \pi \in Q$, $\boldsymbol{\eta} \in V_e$, and $\boldsymbol{\xi} \in V_s$. The variations of (7) are

$$\begin{aligned}
D_{\boldsymbol{\varphi}}\Phi(\boldsymbol{\psi}) &= \int_B \bar{p} \frac{\partial \theta}{\partial \boldsymbol{\varphi}}[\boldsymbol{\psi}] \, dV + \int_B \bar{\mathbf{s}} : \frac{\partial \mathbf{e}}{\partial \boldsymbol{\varphi}}[\boldsymbol{\psi}] \, dV \\
&\quad - \int_B R\mathbf{B} \cdot \boldsymbol{\psi} \, dV - \int_{\partial_T B} \mathbf{T} \cdot \boldsymbol{\psi} \, dS = 0, \\
D_{\bar{\theta}}\Phi(\vartheta) &= \int_B \left(\frac{\partial W^{\text{vol}}}{\partial \bar{\theta}} - \bar{p} \right) \vartheta \, dV = 0, \\
D_{\bar{\mathbf{e}}}\Phi(\boldsymbol{\eta}) &= \int_B \left(\frac{\partial W^{\text{dev}}}{\partial \bar{\mathbf{e}}} - \bar{\mathbf{s}} \right) : \boldsymbol{\eta} \, dV = 0, \\
D_{\bar{p}}\Phi(\pi) &= \int_B (\theta - \bar{\theta}) \pi \, dV = 0, \\
D_{\bar{\mathbf{s}}}\Phi(\boldsymbol{\xi}) &= \int_B (\mathbf{e} - \bar{\mathbf{e}}) : \boldsymbol{\xi} \, dV = 0.
\end{aligned} \tag{8}$$

The two kinematic directional derivatives appearing in (8)₁ admit the explicit forms

$$\frac{\partial \theta}{\partial \boldsymbol{\varphi}}[\boldsymbol{\psi}] = \mathbf{F}^{-\text{T}} : \text{Grad } \boldsymbol{\psi}, \quad \frac{\partial \mathbf{e}}{\partial \boldsymbol{\varphi}}[\boldsymbol{\psi}] = \mathbb{L}^{\text{dev}}(\mathbf{C}) : \text{sym}(\mathbf{F}^{\text{T}} \text{Grad } \boldsymbol{\psi}), \tag{9}$$

where $\mathbb{L}(\mathbf{C}) := \partial \log \mathbf{C} / \partial \mathbf{C}$ is the fourth-order Fréchet derivative of the matrix logarithm and $\mathbb{L}^{\text{dev}}(\mathbf{C}) := \mathbb{L}(\mathbf{C}) - \frac{1}{3} \mathbf{I} \otimes \mathbf{C}^{-1}$ its deviatoric part. The volumetric expression is algorithm-free thanks to Jacobi's formula, while the deviatoric one requires a dedicated evaluation of $\mathbb{L}(\mathbf{C})$; the derivation and a quadrature-point evaluation strategy are given in A.

The corresponding Euler-Lagrange equations are

$$\begin{aligned}
\text{Div } \mathbf{P} + R\mathbf{B} &= \mathbf{0} \quad \text{in } B, \quad \mathbf{P}\mathbf{N} = \mathbf{T} \quad \text{on } \partial_T B, \\
\bar{p} &= \partial W^{\text{vol}} / \partial \bar{\theta} = \kappa \bar{\theta} \quad \text{in } B, \\
\bar{\mathbf{s}} &= \partial W^{\text{dev}} / \partial \bar{\mathbf{e}} = 2\mu \bar{\mathbf{e}} \quad \text{in } B, \\
\bar{\theta} &= \log J \quad \text{in } B, \\
\bar{\mathbf{e}} &= \text{dev} \left[\frac{1}{2} \log(\mathbf{F}^{\text{T}} \mathbf{F}) \right] \quad \text{in } B,
\end{aligned} \tag{10}$$

where $\mathbf{P}$ is the first Piola-Kirchhoff stress reconstructed from the independent fields as

$$\mathbf{P} = (\bar{p}\mathbf{I} + \bar{\mathbf{s}}) \mathbf{F}^{-\text{T}} \quad (\text{equivalently } \mathbf{P} = \boldsymbol{\tau} \mathbf{F}^{-\text{T}} \text{ with } \boldsymbol{\tau} = \bar{p}\mathbf{I} + \bar{\mathbf{s}}), \tag{11}$$

with $\boldsymbol{\tau} = J\boldsymbol{\sigma}$ the spatial Kirchhoff stress. Equivalently, in a formulation organized around the Kirchhoff measure, $\mathbf{P}$ is obtained by the push-forward appropriate to the chosen work-conjugate pair.

3 Discretization

3.1 Finite Element Spaces

Introduce discretizations on a simplicial mesh of B :

$$\begin{aligned}
 \boldsymbol{\varphi}(\mathbf{X}) &:= N_a(\mathbf{X}) \boldsymbol{\varphi}_a \in U_h \subset U, \\
 \bar{\boldsymbol{\theta}}(\mathbf{X}) &:= \mu_m(\mathbf{X}) \bar{\boldsymbol{\theta}}_m \in Q_{\theta,h} \subset Q, \\
 \bar{\mathbf{e}}(\mathbf{X}) &:= \boldsymbol{\lambda}_\alpha^e(\mathbf{X}) \bar{\mathbf{e}}_\alpha \in V_{e,h} \subset V_e, \\
 \bar{p}(\mathbf{X}) &:= \mu_n(\mathbf{X}) \bar{p}_n \in Q_{p,h} \subset Q, \\
 \bar{\mathbf{s}}(\mathbf{X}) &:= \boldsymbol{\lambda}_\beta^s(\mathbf{X}) \bar{\mathbf{s}}_\beta \in V_{s,h} \subset V_s.
 \end{aligned} \tag{12}$$

At polynomial degree $k \geq 2$, a stable choice on tetrahedra is:

- N_a : continuous barycentric polynomial of degree k on the tetrahedron for the displacement;
- μ_m, μ_n : discontinuous L^2 scalar of degree $k - 1$ for $\bar{\boldsymbol{\theta}}$ and $\bar{p}$;
- $\boldsymbol{\lambda}_\alpha^e$: symmetric-tensor-valued basis for $H(\text{Curl}; \mathbb{S}^{\text{dev}})$ of degree $k - 1$ (e.g. Nédélec-type rows with symmetry and trace-free constraints enforced algebraically);
- $\boldsymbol{\lambda}_\beta^s$: symmetric-tensor-valued basis for $H(\text{Div}; \mathbb{S}^{\text{dev}})$ of degree $k - 1$, using the weakly-symmetric construction of Arnold et al. [3] with a skew multiplier in $L^2(B; \mathbb{K})$, or the strongly-symmetric construction of Arnold et al. [4].

Under these choices the discrete formulation satisfies the relevant inf-sup stability conditions for the coupled Hu-Washizu system, with constants independent of the bulk-to-shear ratio up to the range $\kappa/\mu \lesssim 10^4$ relevant to the target problem class. Full incompressibility ($\kappa \rightarrow \infty$) is outside the scope of the present formulation.

Two aspects of the space choice deserve comment: why $\bar{\boldsymbol{\theta}}$ and $\bar{p}$ are *allowed* to be discontinuous while $\bar{\mathbf{e}}$ and $\bar{\mathbf{s}}$ are not, and why the polynomial degrees in the four auxiliary slots are forced to $k - 1$.

Integrability versus stability. The Sobolev regularity required of each auxiliary field must be addressed in two stages. First, the HW functional (7) and its variations (8) must be well-defined: none of the four auxiliary fields appears under a spatial derivative, so $L^2(B)$ is sufficient for integrability of every term (the volumetric energy $W^{\text{vol}}(\bar{\boldsymbol{\theta}})$ depends on values only; the Lagrange pairings $\int \bar{p}(\boldsymbol{\theta} - \bar{\boldsymbol{\theta}})$ and $\int \bar{\mathbf{s}} : (\mathbf{e} - \bar{\mathbf{e}})$ carry no derivatives on the auxiliary fields; and in the momentum variation (8)₁ the gradient acts on the test function $\boldsymbol{\psi}$, not on $\bar{p}$ or $\bar{\mathbf{s}}$). Integrability alone would therefore admit L^2 for all four. Second, however, the coupled saddle-point system must be *inf-sup stable* uniformly in h and in κ/μ , and stability is a stronger requirement than integrability.

Stability is inherited from the symmetric elasticity complex

$$H^1(B; \mathbb{R}^3) \xrightarrow{\text{sym Grad}} H(\text{Curl}; B; \mathbb{S}) \xrightarrow{\text{inc}} H(\text{Div}; B; \mathbb{S}) \xrightarrow{\text{Div}} L^2(B; \mathbb{R}^3) \tag{13}$$

[2, 5], whose discrete counterpart admits cochain projections and a commuting diagram when, and only when, each field sits in the conformity dictated by its slot in the sequence. Consider, for reference, a three-field HW formulation in the primal fields $(\boldsymbol{\varphi}, \bar{\boldsymbol{\epsilon}}, \bar{\boldsymbol{\tau}})$, where $\bar{\boldsymbol{\epsilon}} \in H(\text{Curl}; \mathbb{S})$ and $\bar{\boldsymbol{\tau}} \in H(\text{Div}; \mathbb{S})$ are independent symmetric-tensor fields constrained only weakly to match $\boldsymbol{\epsilon}(\boldsymbol{\varphi})$ and $\boldsymbol{\tau}(\bar{\boldsymbol{\epsilon}})$ through Lagrange-multiplier couplings. Even though $\bar{\boldsymbol{\epsilon}}$ and $\bar{\boldsymbol{\tau}}$ are themselves independent, their volumetric traces $\theta = \frac{1}{3} \text{tr } \bar{\boldsymbol{\epsilon}}$ and $p = \frac{1}{3} \text{tr } \bar{\boldsymbol{\tau}}$ are *embedded scalar components* of globally-coupled tensor fields: the tangential-continuity requirement on $\bar{\boldsymbol{\epsilon}}$ and the normal-continuity requirement on $\bar{\boldsymbol{\tau}}$ prevent these scalar components from being chosen independently in discontinuous L^2 .

In the five-field decomposition, by contrast, $\bar{\boldsymbol{\theta}}, \bar{\boldsymbol{e}}, \bar{p}, \bar{\boldsymbol{s}}$ are introduced as *independent* auxiliary fields, related to the trace and trace-free parts of $\boldsymbol{\epsilon}$ and $\boldsymbol{\tau}$ only weakly through the Lagrange-multiplier constraints. Peeling the trace parts out of the tensor fields severs their tensorial coupling: each auxiliary field is free to sit in whatever space is compatible with its own weak form, subject only to the inf-sup requirement of the coupled system. The stability of the coupled system then factors along the orthogonal volumetric–deviatoric split. The deviatoric part $(\boldsymbol{\varphi}, \bar{\boldsymbol{e}}, \bar{\boldsymbol{s}})$ is governed by a symmetric elasticity complex restricted to the trace-free subspace $\mathbb{S}^{\text{dev}}$, which dictates $\bar{\boldsymbol{e}} \in H(\text{Curl}; \mathbb{S}^{\text{dev}})$ and $\bar{\boldsymbol{s}} \in H(\text{Div}; \mathbb{S}^{\text{dev}})$; here L^2 conformity is insufficient and admits spurious stress modes (the symmetric-tensor analog of P_1 – P_0 checkerboarding). The volumetric part $(\boldsymbol{\varphi}, \bar{\boldsymbol{\theta}}, \bar{p})$ is governed by a Stokes-type mixed sub-problem (detailed below), whose stability is classical and is attained with discontinuous L^2 scalars of degree $k - 1$. Discontinuous L^2 bases are therefore admissible for $\bar{\boldsymbol{\theta}}$ and $\bar{p}$, and inadmissible for $\bar{\boldsymbol{e}}$ and $\bar{\boldsymbol{s}}$: not because the scalars stand outside the complex algebraically, but because treating them as independent fields in a Stokes-like sub-problem is what permits, and is then stabilized by, the weaker L^2 conformity.

The volumetric sub-problem as a Stokes-type system. By “Stokes-type mixed sub-problem” we mean a saddle-point problem with a vector primal field in H^1 , a scalar multiplier in L^2 , and an inf-sup pairing of the form $\int q \mathcal{D}v$ where $\mathcal{D}$ is a first-order scalar differential operator on the primal. The classical Stokes problem

$$-\mu \Delta u + \text{Grad } p = f, \quad \text{Div } u = 0, \quad (14)$$

is the prototype: $u \in H^1(B; \mathbb{R}^3)$, $p \in L^2(B)$, and p enforces $\text{Div } u = 0$ through the pairing $\int p \text{Div } v$. Its stability is the Babuška–Brezzi condition

$$\inf_{q \in L^2} \sup_{v \in H^1} \frac{\int_B q \text{Div } v \, dV}{\|v\|_{H^1} \|q\|_{L^2}} \geq \beta > 0, \quad (15)$$

and the discrete pairs satisfying its analog uniformly in h (Taylor–Hood P_k – P_{k-1} continuous, P_k – P_{k-1} discontinuous, MINI, Crouzeix–Raviart) are classical.

The volumetric sub-problem $(\boldsymbol{\varphi}, \bar{\boldsymbol{\theta}}, \bar{p})$ has the same skeleton, with three substitutions summarized in Table 1.

At $\boldsymbol{F} = \boldsymbol{I}$ the pairing reduces to the classical Stokes pairing via $\boldsymbol{F}^{-\text{T}}: \text{Grad } \boldsymbol{\psi} \rightarrow \text{Div } \boldsymbol{\psi}$, so the two problems coincide in the small-strain limit. Away from the reference configuration, the pairing is the nonlinear finite-deformation generalization, but the saddle-point

Classical Stokes	Volumetric sub-problem
velocity $u \in H^1(B; \mathbb{R}^3)$	$\boldsymbol{\varphi} \in H^1(B; \mathbb{R}^3)$
pressure $p \in L^2(B)$	$\bar{p} \in L^2(B)$
constraint $\text{Div } u = 0$	$\log J - \bar{\theta} = 0$ (via $\bar{p}$), $\kappa \bar{\theta} - \bar{p} = 0$ (via $\bar{\theta}$)
pairing $\int p \text{Div } v$	$\int \bar{p} \mathbf{F}^{-T} : \text{Grad } \boldsymbol{\psi} = \int \bar{p} \partial \theta / \partial \boldsymbol{\varphi} [\boldsymbol{\psi}]$

Table 1: Correspondence between the classical Stokes mixed problem and the volumetric sub-problem of the SEC5L element.

structure, the role of $\bar{p}$ as scalar multiplier, and the inf-sup stability requirement are the same. Consequently the stable discrete pairs known for Stokes lift directly to the volumetric sub-problem: in particular, the P_k - P_{k-1} discontinuous pair is inf-sup stable (uniformly in h , with a κ/μ dependence bounded on the target range) and justifies the discontinuous L^2 choice for $\bar{\theta}$ and $\bar{p}$ at degree $k - 1$.

No comparable classical result exists for the deviatoric sub-problem $(\boldsymbol{\varphi}, \bar{\mathbf{e}}, \bar{\mathbf{s}})$: its stability relies on the AFW / Hu–Zhang finite-element exterior calculus for the symmetric elasticity complex, which is what forces the stronger $H(\text{Curl}; \mathbb{S}^{\text{dev}})$ and $H(\text{Div}; \mathbb{S}^{\text{dev}})$ conformities there.

Determination of the polynomial degrees. The degrees are not free either: on trimmed (Whitney / first-kind Nédélec) spaces each operator of the complex (13) drops one polynomial degree, so selecting continuous barycentric polynomials of degree k for $\boldsymbol{\varphi}$ forces the four auxiliary slots to degree $k - 1$:

$$\underbrace{k}_{\boldsymbol{\varphi}} \xrightarrow{\text{sym Grad}} \underbrace{k-1}_{\bar{\mathbf{e}}} \xrightarrow{\text{inc}} \underbrace{k-1}_{\bar{\mathbf{s}}} \xrightarrow{\text{Div}} \underbrace{k-1}_{\bar{p}, \bar{\theta}}. \quad (16)$$

The scalar slots $\bar{\theta}, \bar{p}$ acquire degree $k - 1$ because they are dual to, respectively, the trace of $\text{Grad } \boldsymbol{\varphi}$ (which is a degree- $(k - 1)$ scalar) and itself. The choice $k = 2$ is the lowest robust one: $k = 1$ triggers the P_1 - P_0 analog of the Stokes counterexample on tetrahedra and is generically inf-sup-unstable for mixed elasticity without enrichment, whereas $k = 2$ yields linear-discontinuous scalar fields for $\bar{\theta}$ and $\bar{p}$ with inf-sup constants bounded independently of h and of κ/μ on the range above.

3.2 Discrete Residuals

Substituting (12) into (8) yields five discrete residual equations:

$$\begin{aligned}
\mathbf{R}_{\boldsymbol{\varphi};a} &:= \int_B \bar{p} \frac{\partial \theta}{\partial \boldsymbol{\varphi}_a} dV + \int_B \bar{\mathbf{s}} : \frac{\partial \mathbf{e}}{\partial \boldsymbol{\varphi}_a} dV \\
&\quad - \int_B R \mathbf{B} N_a dV - \int_{\partial_T B} \mathbf{T} N_a dS = \mathbf{0}, \\
R_{\bar{\theta};m} &:= \int_B \mu_m (\kappa \bar{\theta} - \bar{p}) dV = 0, \\
\mathbf{R}_{\bar{\mathbf{e}};\alpha} &:= \int_B \boldsymbol{\lambda}_\alpha^e : (2\mu \bar{\mathbf{e}} - \bar{\mathbf{s}}) dV = \mathbf{0}, \\
R_{\bar{p};n} &:= \int_B \mu_n (\theta(\boldsymbol{\varphi}) - \bar{\theta}) dV = 0, \\
\mathbf{R}_{\bar{\mathbf{s}};\beta} &:= \int_B \boldsymbol{\lambda}_\beta^s : (\mathbf{e}(\boldsymbol{\varphi}) - \bar{\mathbf{e}}) dV = \mathbf{0}.
\end{aligned} \tag{17}$$

The kinematic derivatives in (17)₁ follow from (9) by taking $\boldsymbol{\psi} = N_a \boldsymbol{\psi}_a$ and isolating the DOF direction $\boldsymbol{\psi}_a$:

$$\frac{\partial \theta}{\partial \boldsymbol{\varphi}_a} = \mathbf{F}^{-T} \text{Grad } N_a, \quad \frac{\partial \mathbf{e}}{\partial \boldsymbol{\varphi}_a} = \mathbb{L}^{\text{dev}}(\mathbf{C}) : \text{sym}(\mathbf{F} \otimes^s \text{Grad } N_a), \tag{18}$$

where $\mathbf{F}^{-T} \text{Grad } N_a \in \mathbb{R}^3$ is the volumetric B-vector and $(\mathbf{F} \otimes^s \text{Grad } N_a)_{ijk} := \frac{1}{2} (F_{ij} N_{a,k} + F_{ik} N_{a,j}) \in \mathbb{S} \otimes \mathbb{R}^3$ is the deviatoric B-tensor, both local to node a and evaluated quadrature-point-wise.

3.3 Projection Structure

The four auxiliary equations (17)₂₋₅ are each L^2 projections from pointwise quantities onto the corresponding finite element subspace. Define the constant mass operators in the reference configuration:

$$\begin{aligned}
[M_\theta]_{mn} &:= \int_B \mu_m \mu_n dV, & [M_e]_{\alpha\beta} &:= \int_B \boldsymbol{\lambda}_\alpha^e : \boldsymbol{\lambda}_\beta^e dV, \\
[M_p]_{mn} &:= \int_B \mu_m \mu_n dV, & [M_s]_{\alpha\beta} &:= \int_B \boldsymbol{\lambda}_\alpha^s : \boldsymbol{\lambda}_\beta^s dV.
\end{aligned} \tag{19}$$

The discrete auxiliary fields are recovered from $\boldsymbol{\varphi}$ by solving four linear projections with pre-factored constant SPD matrices:

$$\begin{aligned}
M_\theta \cdot \{\bar{\theta}\}_{\text{dof}} &= \int_B \mu_m \log J(\boldsymbol{\varphi}) dV, \\
M_e \cdot \{\bar{\mathbf{e}}\}_{\text{dof}} &= \int_B \boldsymbol{\lambda}_\alpha^e : \mathbf{e}(\boldsymbol{\varphi}) dV, \\
M_p \cdot \{\bar{p}\}_{\text{dof}} &= \int_B \mu_m \kappa \log J(\boldsymbol{\varphi}) dV, \\
M_s \cdot \{\bar{\mathbf{s}}\}_{\text{dof}} &= \int_B \boldsymbol{\lambda}_\alpha^s : [2\mu \mathbf{e}(\boldsymbol{\varphi}) - 2\mu \mathbf{e}^p] dV.
\end{aligned} \tag{20}$$

In a total-Lagrangian setting the mass matrices in (19) are independent of $\boldsymbol{\varphi}$ and of time; a single factorization at setup (block-diagonal per element for discontinuous L^2 fields, sparse and globally coupled for conforming $H(\text{Curl})$ and $H(\text{Div})$ fields) serves every Newton iteration of every time step.

Substituting the projected auxiliary fields into (17)₁ collapses the five-field system to a single discrete statement in $\boldsymbol{\varphi}$ alone:

$$\begin{aligned} \mathbf{R}_a(\boldsymbol{\varphi}) &:= \int_B \bar{p}(\boldsymbol{\varphi}) \frac{\partial \theta}{\partial \boldsymbol{\varphi}_a} dV + \int_B \bar{\mathbf{s}}(\boldsymbol{\varphi}) : \frac{\partial \mathbf{e}}{\partial \boldsymbol{\varphi}_a} dV \\ &\quad - \int_B R \mathbf{B} N_a dV - \int_{\partial_T B} \mathbf{T} N_a dS = \mathbf{0}, \end{aligned} \quad (21)$$

in which $\bar{p}(\boldsymbol{\varphi})$ and $\bar{\mathbf{s}}(\boldsymbol{\varphi})$ are understood as the evaluations of the projections (20) applied at the current $\boldsymbol{\varphi}$. The coupling to $\bar{\theta}(\boldsymbol{\varphi})$ and $\bar{\mathbf{e}}(\boldsymbol{\varphi})$ enters implicitly through the plastic update (6) and through the kinematic dependence of $\boldsymbol{\epsilon}(\boldsymbol{\varphi})$ on $\boldsymbol{\varphi}$; in the reduced statement (21) these fields are used only to supply the stress fields needed for equilibrium.

4 Newton Solution

4.1 Linearization

The reduced residual (21) is a nonlinear function of $\boldsymbol{\varphi}$ alone. A Newton iteration computes $\Delta \boldsymbol{\varphi}$ from

$$\mathbf{R}_a(\boldsymbol{\varphi}_n) + \mathbf{K}_{ab} \cdot \Delta \boldsymbol{\varphi}_b = \mathbf{0}, \quad \boldsymbol{\varphi}_{n+1} = \boldsymbol{\varphi}_n + \Delta \boldsymbol{\varphi}, \quad (22)$$

with tangent

$$\mathbf{K}_{ab} := \frac{\partial \mathbf{R}_a}{\partial \boldsymbol{\varphi}_b} = \mathbf{K}_{ab}^{\text{mat}} + \mathbf{K}_{ab}^{\text{geom}}, \quad (23)$$

where the material tangent assembles contributions from both independent stress fields through the chain rule of the projections:

$$\mathbf{K}_{ab}^{\text{mat}} = \int_B \frac{\partial \boldsymbol{\epsilon}(\mathbf{X})}{\partial \boldsymbol{\varphi}_a} : [\kappa \mathcal{P}_p \mathbf{I} \otimes \mathbf{I} + 2\mu \mathcal{P}_s \mathcal{P}_{\text{dev}}] : \frac{\partial \boldsymbol{\epsilon}(\mathbf{X})}{\partial \boldsymbol{\varphi}_b} dV, \quad (24)$$

in which $\mathcal{P}_p$ and $\mathcal{P}_s$ denote the L^2 projection operators onto $Q_{p,h}$ and $V_{s,h}$ respectively, and $\mathcal{P}_{\text{dev}}$ is the algebraic projector onto deviatoric tensors. The geometric tangent $\mathbf{K}_{ab}^{\text{geom}}$ arises from the nonlinear dependence of $\boldsymbol{\epsilon}$ on $\mathbf{F}$ through the matrix logarithm and is obtained analytically via the Fréchet derivative $\mathbb{L}(\mathbf{C}) = \partial \log \mathbf{C} / \partial \mathbf{C}$, evaluated by the hybrid algorithm of Ortiz et al. [14].

The effective tangent is pre-factorized at setup (for the parts of the projections that are constant in the reference configuration) and assembled at each Newton iteration. The global nonlinear system solved in (22) is sized by the displacement degrees of freedom only.

4.2 Dynamic Extension

For transient analyses, the momentum balance in (10)₁ is augmented by the inertial term:

$$R \ddot{\boldsymbol{\varphi}} - \text{Div } \mathbf{P} = R \mathbf{B}. \quad (25)$$

The auxiliary fields $\bar{\theta}$, $\bar{\mathbf{e}}$, $\bar{p}$, $\bar{\mathbf{s}}$ carry no time derivatives: their weak statements are algebraic constraints at each instant. Consequently the mass matrix

$$[M_{\boldsymbol{\varphi}}]_{ab} := \int_B R N_a N_b \, dV \quad (26)$$

is concentrated in the displacement block, and the semi-discrete system is a differential-algebraic system of index one. Standard Newmark- β or Hilber-Hughes-Taylor integration applied to the reduced equation (21) with the mass term (26) proceeds identically as in single-field dynamics. The auxiliary-field projections (20) are executed at each quadrature point evaluation of the residual and tangent, using the current $\boldsymbol{\varphi}$.

5 Plasticity Extension

For isotropic J_2 plasticity with isotropic hardening, the Helmholtz free energy (4) is modified to depend on the elastic deviatoric log-strain $\mathbf{e}^e := \bar{\mathbf{e}} - \mathbf{e}^p$ and the equivalent plastic strain $\bar{\epsilon}^p$:

$$A(\bar{\theta}, \bar{\mathbf{e}}, \mathbf{e}^p, \bar{\epsilon}^p) = \frac{\kappa}{2} \bar{\theta}^2 + \mu \|\bar{\mathbf{e}} - \mathbf{e}^p\|^2 + \mathcal{H}(\bar{\epsilon}^p), \quad (27)$$

with $\mathcal{H}$ the hardening potential. The yield function is

$$f(\bar{\mathbf{s}}, \bar{\epsilon}^p) := \|\bar{\mathbf{s}}\| - \sqrt{\frac{2}{3}} [\sigma_y + \mathcal{H}'(\bar{\epsilon}^p)]. \quad (28)$$

The return map is applied at each quadrature point between the kinematic evaluation of $\mathbf{e}(\boldsymbol{\varphi})$ and the projection onto $V_{s,h}$, reproducing the small-strain radial return of Simo [18]:

$$\begin{aligned} \mathbf{e}^{\text{trial}} &:= \mathbf{e}(\boldsymbol{\varphi}) - \mathbf{e}_n^p, \\ \mathbf{s}^{\text{trial}} &:= 2\mu \mathbf{e}^{\text{trial}}, \\ \Delta\gamma &:= \frac{\|\mathbf{s}^{\text{trial}}\| - \sqrt{2/3} [\sigma_y + \mathcal{H}'(\bar{\epsilon}_n^p)]}{2\mu + \frac{2}{3} \mathcal{H}''(\bar{\epsilon}_n^p)} \quad \text{if } f^{\text{trial}} > 0, \quad \Delta\gamma := 0 \text{ otherwise,} \\ \mathbf{n} &:= \mathbf{s}^{\text{trial}} / \|\mathbf{s}^{\text{trial}}\|, \\ \mathbf{e}_{n+1}^p &:= \mathbf{e}_n^p + \Delta\gamma \mathbf{n}, \\ \bar{\epsilon}_{n+1}^p &:= \bar{\epsilon}_n^p + \sqrt{\frac{2}{3}} \Delta\gamma, \\ \mathbf{s}^{\text{pointwise}} &:= 2\mu (\mathbf{e}(\boldsymbol{\varphi}) - \mathbf{e}_{n+1}^p). \end{aligned} \quad (29)$$

The pointwise updated $\mathbf{s}^{\text{pointwise}}$ is then projected onto $V_{s,h}$ via (20)₄. Plastic internal variables $\mathbf{e}^p$ and $\bar{\epsilon}^p$ are stored at quadrature points and evolved pointwise; they are never projected, preserving the exactness of the return map. The Lie-group/log-exp interpolation scheme of Mota et al. [13] applies when the internal variables themselves are tensorial quantities requiring frame-indifferent interpolation across quadrature.

6 Discussion

The SEC5L element inherits three structural properties from its symmetric-elasticity-complex foundation that distinguish it from the composite tetrahedron:

1. **Pointwise discrete symmetry of stress.** The independent stress field $\bar{\mathbf{s}}$ lives in $H(\text{Div}; \mathbb{S}^{\text{dev}})$, so its symmetry is a property of the finite element space rather than of the constitutive law. This eliminates the cross-space aliasing between strain and stress fields that was observed to corrupt the equivalent plastic strain in stabilized displacement formulations.
2. **Normal-traction continuity.** The $H(\text{Div})$ conformity of $\bar{\mathbf{s}}$ ensures that the normal component of the traction field is continuous across element faces. In regimes with large bulk-to-shear ratio, this continuity suppresses pressure checkerboarding that would otherwise develop with discontinuous stress representations.
3. **Algebraically linear constitutive relation.** The Hencky–Kirchhoff pairing renders the constitutive law (5) linear in log-strain space. The J_2 return map (29) is identical in form to the small-strain case of Simo [18]; no iteration is required beyond the closed-form projection. All finite-deformation nonlinearity is confined to the kinematic evaluation $\boldsymbol{\epsilon} = \frac{1}{2} \log(\mathbf{F}^T \mathbf{F})$ and the push-forward in (11).

The projection-and-eliminate architecture combines these structural properties with a linear-algebra cost commensurate with a single-field displacement formulation. Pre-factorization of the four constant mass matrices (19) at setup amortizes their cost over all Newton iterations of all time steps; each iteration incurs only four forward-solves (fast against pre-factored SPD operators) beyond the dominant displacement Newton solve.

Future work will address the implementation of

- symmetric-tensor-valued conforming bases on tetrahedra in the `ReferenceFiniteElements.jl` library [7];
- multi-field assembly with discontinuous L^2 and conforming $H(\text{Curl})$, $H(\text{Div})$ spaces in `FiniteElementContainers.jl` [6];
- verification on canonical problems on which the composite tetrahedron is known to exhibit degraded performance, including a three-dimensional Cook’s membrane at large deformation, contact-driven punch indentation, large-rotation twist of a prismatic bar, and the spherical cavity expansion benchmark appropriate to the high-bulk-modulus regime.

Verification of the inf-sup stability constants on these meshes is a prerequisite for quantitative assessment. Extension to incrementally objective integration of anisotropic plastic response through the multiplicative decomposition $\mathbf{F} = \mathbf{F}^e \mathbf{F}^p$, in which the Hencky–Kirchhoff conjugacy is no longer exact, is deferred to a subsequent formulation on the gradient elasticity complex with independent non-symmetric displacement-gradient and first-Piola-Kirchhoff-stress fields.

A Variations of the Kinematic Relation

This appendix makes explicit the directional derivatives $\partial\theta/\partial\boldsymbol{\varphi}$ and $\partial\mathbf{e}/\partial\boldsymbol{\varphi}$ appearing in the displacement variation (8)₁. The derivation is a chain of elementary steps expressed through the operators tr , vol , and dev , with only one non-elementary ingredient: the fourth-order Fréchet derivative of the matrix logarithm $\mathbb{L}(\mathbf{C}) := \partial \log \mathbf{C} / \partial \mathbf{C}$, whose evaluation by the hybrid Taylor–spectral algorithm of Ortiz et al. [14] is the subject of a separate kernel that is here treated as a primitive.

A.1 Chain of Partial Derivatives

The kinematic relation $\boldsymbol{\epsilon}(\boldsymbol{\varphi}) = \frac{1}{2} \log(\mathbf{F}^T \mathbf{F})$ composes four maps:

$$\boldsymbol{\varphi} \xrightarrow{\text{Grad}} \mathbf{F} \xrightarrow{(\cdot)^T(\cdot)} \mathbf{C} \xrightarrow{\frac{1}{2} \log} \boldsymbol{\epsilon} \xrightarrow{\{\text{tr}, \text{dev}\}} (\theta, \mathbf{e}). \quad (30)$$

Applied to a test function $\boldsymbol{\psi} \in U$ with $\boldsymbol{\psi}|_{\partial\varphi B} = \mathbf{0}$, each link contributes a directional derivative:

$$D_{\boldsymbol{\varphi}} \mathbf{F}[\boldsymbol{\psi}] = \text{Grad } \boldsymbol{\psi}, \quad (31a)$$

$$D_{\mathbf{F}} \mathbf{C}[\text{Grad } \boldsymbol{\psi}] = (\text{Grad } \boldsymbol{\psi})^T \mathbf{F} + \mathbf{F}^T \text{Grad } \boldsymbol{\psi} = 2 \text{sym}(\mathbf{F}^T \text{Grad } \boldsymbol{\psi}), \quad (31b)$$

$$D_{\mathbf{C}} \boldsymbol{\epsilon}[\delta \mathbf{C}] = \frac{1}{2} \mathbb{L}(\mathbf{C}) : \delta \mathbf{C}. \quad (31c)$$

Both minor symmetries of $\mathbb{L}(\mathbf{C})$ follow from symmetry of $\mathbf{C}$; consequently $\mathbb{L}(\mathbf{C}) : A = \mathbb{L}(\mathbf{C}) : \text{sym } A$ for any A .

Composing (31a)–(31c) gives the total directional derivative of the Hencky strain:

$$\frac{\partial \boldsymbol{\epsilon}}{\partial \boldsymbol{\varphi}}[\boldsymbol{\psi}] = \mathbb{L}(\mathbf{C}) : \text{sym}(\mathbf{F}^T \text{Grad } \boldsymbol{\psi}). \quad (32)$$

Its volumetric and deviatoric parts are then obtained by the trace and deviator projections.

A.2 Trace Identity for the Log-Derivative

The volumetric projection of the strain derivative reduces to a classical identity that does not require any eigenvalue information about $\mathbf{C}$. We derive it here in three steps so that the appearance of $\mathbf{C}^{-1}$ (and, through $\mathbf{F}\mathbf{C}^{-1} = \mathbf{F}^{-T}$, of $\mathbf{F}^{-T}$) in the volumetric variation is transparent.

Jacobi's formula. For any invertible second-order tensor $\mathbf{A}$, the derivative of the determinant with respect to its argument is

$$\frac{\partial \det \mathbf{A}}{\partial \mathbf{A}} = (\det \mathbf{A}) \mathbf{A}^{-T}. \quad (33)$$

Applying the chain rule to the composition $\mathbf{A} \mapsto \det \mathbf{A} \mapsto \log \det \mathbf{A}$, the outer derivative $d(\log x)/dx = 1/x$ evaluated at $x = \det \mathbf{A}$ contributes a factor $1/\det \mathbf{A}$ that cancels the $\det \mathbf{A}$ in (33):

$$\frac{\partial \log \det \mathbf{A}}{\partial \mathbf{A}} = \frac{1}{\det \mathbf{A}} \frac{\partial \det \mathbf{A}}{\partial \mathbf{A}} = \frac{1}{\det \mathbf{A}} (\det \mathbf{A}) \mathbf{A}^{-\text{T}} = \mathbf{A}^{-\text{T}}. \quad (34)$$

Specializing to the symmetric positive-definite $\mathbf{C}$, where $\mathbf{C}^{-\text{T}} = \mathbf{C}^{-1}$, gives the form used below:

$$\frac{\partial \log \det \mathbf{C}}{\partial \mathbf{C}} = \mathbf{C}^{-1}. \quad (35)$$

Trace-logarithm identity. For any symmetric positive-definite $\mathbf{C}$,

$$\text{tr} \log \mathbf{C} = \log \det \mathbf{C}. \quad (36)$$

This follows from the spectral representation: if $\mathbf{C} = \sum_i \lambda_i \mathbf{N}_i \otimes \mathbf{N}_i$ with $\lambda_i > 0$, then $\text{tr} \log \mathbf{C} = \sum_i \log \lambda_i = \log \prod_i \lambda_i = \log \det \mathbf{C}$. The identity is independent of eigenvalue multiplicities and thus robust to coalescence.

Contraction of $\mathbb{L}(\mathbf{C})$ with the identity. By definition $\mathbb{L}(\mathbf{C}) = \partial \log \mathbf{C} / \partial \mathbf{C}$, and differentiation commutes with the (constant-coefficient) trace operator, so

$$\mathbf{I} : \mathbb{L}(\mathbf{C}) = \text{tr} \left(\frac{\partial \log \mathbf{C}}{\partial \mathbf{C}} \right) = \frac{\partial \text{tr} \log \mathbf{C}}{\partial \mathbf{C}}. \quad (37)$$

Combining (37) with (36) and (35) yields the trace identity

$$\mathbf{I} : \mathbb{L}(\mathbf{C}) = \frac{\partial \text{tr} \log \mathbf{C}}{\partial \mathbf{C}} = \frac{\partial \log \det \mathbf{C}}{\partial \mathbf{C}} = \mathbf{C}^{-1}. \quad (38)$$

Equation (38) is a structural property of the matrix logarithm: it holds pointwise for every symmetric positive-definite $\mathbf{C}$, independently of the particular algorithm used to evaluate $\mathbb{L}(\mathbf{C})$. In particular, the hybrid Taylor–spectral scheme of Ortiz et al. [14] satisfies this identity exactly in both of its branches, so the volumetric part of the variation is always recovered in closed form as $\mathbf{C}^{-1}$ and does not inherit any of the ill-conditioning associated with eigenvalue coalescence.

A.3 Volumetric Part

Applying tr to (32) and using (38),

$$\begin{aligned} \frac{\partial \theta}{\partial \boldsymbol{\varphi}}[\boldsymbol{\psi}] &= \text{tr} \left\{ \frac{\partial \boldsymbol{\epsilon}}{\partial \boldsymbol{\varphi}}[\boldsymbol{\psi}] \right\} = [\mathbf{I} : \mathbb{L}(\mathbf{C})] : \text{sym}(\mathbf{F}^{\text{T}} \text{Grad } \boldsymbol{\psi}) \\ &= \mathbf{C}^{-1} : \text{sym}(\mathbf{F}^{\text{T}} \text{Grad } \boldsymbol{\psi}) = \mathbf{C}^{-1} : \mathbf{F}^{\text{T}} \text{Grad } \boldsymbol{\psi} = \mathbf{F}^{-\text{T}} : \text{Grad } \boldsymbol{\psi}, \end{aligned} \quad (39)$$

where the penultimate step drops the sym because $\mathbf{C}^{-1}$ is itself symmetric. The last equality follows from moving $\mathbf{F}^T$ across the double contraction and then simplifying $\mathbf{F}\mathbf{C}^{-1}$. Using the standard identity $\mathbf{A} : (\mathbf{B}\mathbf{C}) = (\mathbf{B}^T \mathbf{A}) : \mathbf{C}$ with $\mathbf{A} = \mathbf{C}^{-1}$, $\mathbf{B} = \mathbf{F}^T$, and $\mathbf{C} = \text{Grad } \boldsymbol{\psi}$,

$$\mathbf{C}^{-1} : \mathbf{F}^T \text{Grad } \boldsymbol{\psi} = (\mathbf{F}\mathbf{C}^{-1}) : \text{Grad } \boldsymbol{\psi}, \quad (40)$$

and the product $\mathbf{F}\mathbf{C}^{-1}$ collapses by direct algebra:

$$\mathbf{F}\mathbf{C}^{-1} = \mathbf{F} (\mathbf{F}^T \mathbf{F})^{-1} = \mathbf{F} \mathbf{F}^{-1} \mathbf{F}^{-T} = \mathbf{F}^{-T}. \quad (41)$$

Substituting (41) into (40) gives the last equality in (39). The volumetric tensor part of the strain variation is then $\text{vol}(\delta\boldsymbol{\epsilon}) = \frac{1}{3} \delta\theta \mathbf{I}$.

Equation (39) recovers the classical identity $\partial \log J / \partial \mathbf{F} = \mathbf{F}^{-T}$ from within the log-strain framework. In the reduced residual, this means that the pressure term $\int_B \bar{p} (\partial\theta / \partial \boldsymbol{\varphi})$ reduces to $\int_B \bar{p} \mathbf{F}^{-T} : \text{Grad } \boldsymbol{\psi}$, a form that requires neither the full $\mathbb{L}(\mathbf{C})$ nor its spectral decomposition.

A.4 Deviatoric Part

Applying $\text{dev} = \text{id} - \text{vol}$ to (32),

$$\begin{aligned} \frac{\partial \mathbf{e}}{\partial \boldsymbol{\varphi}}[\boldsymbol{\psi}] &= \text{dev} \left\{ \frac{\partial \boldsymbol{\epsilon}}{\partial \boldsymbol{\varphi}}[\boldsymbol{\psi}] \right\} = \frac{\partial \boldsymbol{\epsilon}}{\partial \boldsymbol{\varphi}}[\boldsymbol{\psi}] - \frac{1}{3} \frac{\partial \theta}{\partial \boldsymbol{\varphi}}[\boldsymbol{\psi}] \mathbf{I} \\ &= \mathbb{L}(\mathbf{C}) : \text{sym}(\mathbf{F}^T \text{Grad } \boldsymbol{\psi}) - \frac{1}{3} (\mathbf{F}^{-T} : \text{Grad } \boldsymbol{\psi}) \mathbf{I}. \end{aligned} \quad (42)$$

Equivalently, introducing the deviatoric log-derivative

$$\mathbb{L}^{\text{dev}}(\mathbf{C}) := \mathbb{L}(\mathbf{C}) - \frac{1}{3} \mathbf{I} \otimes \mathbf{C}^{-1}, \quad (43)$$

the deviatoric variation has the one-line form

$$\frac{\partial \mathbf{e}}{\partial \boldsymbol{\varphi}}[\boldsymbol{\psi}] = \mathbb{L}^{\text{dev}}(\mathbf{C}) : \text{sym}(\mathbf{F}^T \text{Grad } \boldsymbol{\psi}). \quad (44)$$

By construction $\mathbf{I} : \mathbb{L}^{\text{dev}}(\mathbf{C}) = \mathbf{0}$, so the range of (44) is trace-free, matching the space $\mathbb{S}^{\text{dev}}$ of $\mathbf{e}$.

No further analytical reduction. The contrast between the volumetric and deviatoric branches is structural. In A.3, the fourth-order $\mathbb{L}(\mathbf{C})$ never appears in the final expression: the volumetric projection is the contraction $\mathbf{I} : \mathbb{L}(\mathbf{C})$, and that contraction coincides with $\partial \log \det \mathbf{C} / \partial \mathbf{C}$, which has the closed form $\mathbf{C}^{-1}$ by Jacobi's formula (34). The volumetric part thus bypasses the eigenstructure of $\mathbf{C}$ entirely. In (44), by contrast, the deviatoric projection only removes from $\mathbb{L}(\mathbf{C})$ the single piece already known in closed form; the remainder of $\mathbb{L}(\mathbf{C})$ encodes the full fourth-order information of the matrix-logarithm derivative and has no closed form in general, since it depends on the divided differences $(\log \lambda_i - \log \lambda_j) / (\lambda_i - \lambda_j)$

of the eigenvalues of $\mathbf{C}$. The deviatoric branch therefore always requires a dedicated evaluation of $\mathbb{L}(\mathbf{C})$ — e.g. the hybrid Taylor–spectral algorithm of Ortiz et al. [14] — and is the branch that inherits the eigenvalue-coalescence issue. This asymmetry motivates handling the volumetric and deviatoric parts with different quadrature-point routines rather than a single monolithic evaluation of $\mathbb{L}(\mathbf{C})$.

An eigenvalue-free alternative is the Padé–Fréchet algorithm of Al-Mohy et al. [1], which extends the inverse scaling-and-squaring Padé framework already invoked in (1) for $\log \mathbf{C}$ to produce the Fréchet derivative $L_{\log}(\mathbf{C}, \mathbf{E})$ in the same sweep, propagating the direction $\mathbf{E}$ through the Sylvester equations of the square-root chain and through the Padé kernel. Applied to each of the six symmetric basis directions on $\mathbb{S}$, the algorithm assembles the full fourth-order $\mathbb{L}(\mathbf{C})$ without any eigendecomposition, removing the coalescence branch entirely and unifying the numerical treatment of $\log \mathbf{C}$ and $\partial \log \mathbf{C} / \partial \mathbf{C}$ under a single kernel.

A.5 Computational Summary

At each quadrature point of the residual assembly (17)₁ the evaluation proceeds in five steps:

1. Kinematics: $\mathbf{F} = \text{Grad } \varphi_h$, $\mathbf{C} = \mathbf{F}^T \mathbf{F}$, $\mathbf{F}^{-T}$.
2. (Optionally) spectral decomposition of $\mathbf{C}$, skipped when $\|\mathbf{C} - \mathbf{I}\|$ falls below the Taylor threshold of Ortiz et al. [14].
3. Evaluation of $\mathbb{L}(\mathbf{C})$ by the hybrid algorithm (spectral branch or Taylor recurrence).
4. For each displacement basis N_a and direction $\mathbf{e}_i$, with $G_{a,i} := \text{Grad}(N_a \mathbf{e}_i)$:

$$\begin{aligned} \delta \theta_{a,i} &= \mathbf{F}^{-T} : G_{a,i}, \\ \delta \boldsymbol{\epsilon}_{a,i} &= \mathbb{L}(\mathbf{C}) : \text{sym}(\mathbf{F}^T G_{a,i}), \\ \delta \mathbf{e}_{a,i} &= \delta \boldsymbol{\epsilon}_{a,i} - \frac{1}{3} \delta \theta_{a,i} \mathbf{I}. \end{aligned}$$

5. Contraction with the stress fields: $(\bar{p} \mathbf{I} + \bar{\mathbf{s}}) : \delta \boldsymbol{\epsilon}_{a,i} = \bar{p} \delta \theta_{a,i} + \bar{\mathbf{s}} : \delta \mathbf{e}_{a,i}$, consistent with the orthogonal split (2)–(3).

The volumetric and deviatoric variations share a single evaluation of $\mathbb{L}(\mathbf{C})$; no separate constitutive-level call is required for the trace, which is obtained from $\mathbf{F}^{-T}$ alone. The only object that depends on the kinematic state at the current Newton iterate is $\mathbb{L}(\mathbf{C})$ itself.

B The Symmetric Elasticity Complex

This appendix collects the structural background behind the complex (13). The goal is operational: to identify each operator, its kernel, its range, and its interaction with the orthogonal volumetric–deviatoric split $\mathbb{S} = \mathbb{S}^{\text{vol}} \oplus \mathbb{S}^{\text{dev}}$ that organizes the five-field formulation.

B.1 The de Rham Complex and its Tensorial Lift

On a bounded, simply-connected, Lipschitz domain $B \subset \mathbb{R}^3$, the scalar de Rham complex

$$\mathbb{R} \xrightarrow{\iota} H^1(B) \xrightarrow{\text{Grad}} H(\text{Curl}; B) \xrightarrow{\text{Curl}} H(\text{Div}; B) \xrightarrow{\text{Div}} L^2(B) \xrightarrow{0} 0 \quad (45)$$

is exact: $\text{Curl} \circ \text{Grad} = 0$, $\text{Div} \circ \text{Curl} = 0$, and the kernel of each operator coincides with the range of its predecessor. Exactness encodes the classical vector-calculus identities (“Curl-free implies gradient,” “Div-free implies curl”) as statements about the kernels and ranges of differential operators between Sobolev spaces.

The symmetric elasticity complex is the symmetric-tensor analog of (45). The Bernstein–Gel’fand–Gel’fand (BGG) construction of Arnold et al. [2, 5] couples a scalar de Rham complex with a vector-valued one through an algebraic projection, resolving the resulting double complex to yield

$$\text{RM}(B) \xrightarrow{\iota} H^1(B; \mathbb{R}^3) \xrightarrow{\text{sym Grad}} H(\text{Curl}; B; \mathbb{S}) \xrightarrow{\text{inc}} H(\text{Div}; B; \mathbb{S}) \xrightarrow{\text{Div}} L^2(B; \mathbb{R}^3) \xrightarrow{0} 0, \quad (46)$$

where $\text{RM}(B) = \{\mathbf{a} + \mathbf{W}\mathbf{X} : \mathbf{a} \in \mathbb{R}^3, \mathbf{W} \in \text{skew}(\mathbb{R}^3)\}$ is the six-dimensional space of infinitesimal rigid motions. Each space in (46) carries a symmetric-tensor (or vector) field with Sobolev conformity dictated by the operator it must survive; the complex is exact on a simply-connected domain.

B.2 Operators, Kernels, and Ranges

The three differential operators of (46) are

1. **Symmetric gradient.** $\text{sym Grad} : H^1(B; \mathbb{R}^3) \rightarrow L^2(B; \mathbb{S})$ with

$$\text{sym Grad } \mathbf{u} := \frac{1}{2}(\text{Grad } \mathbf{u} + \text{Grad } \mathbf{u}^T). \quad (47)$$

Its kernel is $\text{RM}(B)$, the space of infinitesimal rigid motions; this is the Korn–Kirchhoff theorem.

2. **Incompatibility.** $\text{inc} : H(\text{Curl}; B; \mathbb{S}) \rightarrow H(\text{Div}; B; \mathbb{S})$ is the second-order operator

$$\text{inc}(\boldsymbol{\epsilon}) := \text{Curl}(\text{Curl } \boldsymbol{\epsilon})^T, \quad [\text{inc}(\boldsymbol{\epsilon})]_{pq} = \varepsilon_{pij} \varepsilon_{qkl} \epsilon_{jl,ik}, \quad (48)$$

where the Curl of a tensor field acts row-wise and ε_{ijk} is the Levi-Civita symbol. The kernel of inc coincides with the range of sym Grad : a symmetric-tensor field $\boldsymbol{\epsilon}$ is compatible with a displacement, $\boldsymbol{\epsilon} = \text{sym Grad } \mathbf{u}$, if and only if $\text{inc}(\boldsymbol{\epsilon}) = 0$. This is the classical *Saint-Venant compatibility condition*.

3. **Divergence.** $\text{Div} : H(\text{Div}; B; \mathbb{S}) \rightarrow L^2(B; \mathbb{R}^3)$ acts row-wise, $(\text{Div } \boldsymbol{\tau})_i = \tau_{ij,j}$. Its kernel is the range of inc : a symmetric-tensor stress field is self-equilibrated if and only if it is the inc of a symmetric-tensor strain field.

The complex loses one order of regularity at each step ($H^1 \rightarrow H(\text{Curl}) \rightarrow H(\text{Div}) \rightarrow L^2$) and is exact on a simply-connected domain. The exactness at the L^2 slot is the solvability of the equilibrium equation $\text{Div } \boldsymbol{\tau} = -R\mathbf{B}$ in $H(\text{Div}; \mathbb{S})$ for any admissible body-force data.

B.3 Interaction with the Volumetric–Deviatoric Split

The orthogonal decomposition $\mathbb{S} = \mathbb{S}^{\text{vol}} \oplus \mathbb{S}^{\text{dev}}$ does *not* descend to a pair of sub-complexes of (46). The middle operator inc mixes volumetric and deviatoric parts, as the following explicit computation shows.

Take a purely volumetric input $\boldsymbol{\epsilon} = \alpha \mathbf{I}$ with $\alpha \in H^1(B)$ and $\mathbf{I}$ the second-order identity. We regard $\alpha \mathbf{I}$ as an element of $H(\text{Curl}; \mathbb{S})$, not as a compatible strain: for generic non-affine α the field $\alpha \mathbf{I}$ lies outside the range of sym Grad , since $\text{inc}(\alpha \mathbf{I}) = 0$ forces $\alpha_{,pq} = 0$ for $p \neq q$ and $\Delta \alpha = \alpha_{,pp}$ (no sum), whose only solutions are affine $\alpha = a + b_i X_i$. The counterexample below is informative precisely because it probes inc on the non-compatible part of the middle space. Substituting $\epsilon_{jl} = \alpha \delta_{jl}$ into (48) and contracting with the Levi-Civita identity $\varepsilon_{pij} \varepsilon_{qkj} = \delta_{pq} \delta_{ik} - \delta_{pk} \delta_{iq}$,

$$[\text{inc}(\alpha \mathbf{I})]_{pq} = \varepsilon_{pij} \varepsilon_{qki} \alpha_{,jk} = \delta_{pq} \alpha_{,ii} - \alpha_{,pq}, \quad (49)$$

or in coordinate-free form,

$$\text{inc}(\alpha \mathbf{I}) = (\Delta \alpha) \mathbf{I} - \text{Grad Grad } \alpha. \quad (50)$$

Projecting onto the vol–dev split,

$$\text{inc}(\alpha \mathbf{I}) = \underbrace{\frac{2}{3}(\Delta \alpha) \mathbf{I}}_{\text{vol}} - \underbrace{\left[\text{Grad Grad } \alpha - \frac{1}{3}(\Delta \alpha) \mathbf{I} \right]}_{\text{dev}}, \quad (51)$$

where the deviatoric part is the trace-free Hessian of α , generically nonzero. A purely volumetric strain is therefore mapped by inc to a stress with nonzero deviatoric part. The converse also fails: the trace of $\text{inc } \boldsymbol{\epsilon}$ is $\text{tr}[\text{inc } \boldsymbol{\epsilon}] = \epsilon_{ij,ij}$, which is nonzero for a generic deviatoric $\boldsymbol{\epsilon} \in \mathbb{S}^{\text{dev}}$.

This is the structural reason why the volumetric and deviatoric sub-problems of the five-field formulation cannot be decoupled at the level of the complex itself: the complex-compatible conformities $H(\text{Curl}; \mathbb{S})$ and $H(\text{Div}; \mathbb{S})$ act on full symmetric tensors, and the trace-free restrictions inherit their stability from the full-tensor theory rather than from an independent sub-complex result. What the vol–dev split *does* provide is an orthogonal decomposition of the constitutive operator through its action on a strain field,

$$\mathbb{C} : \boldsymbol{\epsilon} = \kappa (\text{tr } \boldsymbol{\epsilon}) \mathbf{I} + 2\mu \text{ dev } \boldsymbol{\epsilon}, \quad (52)$$

which decouples the bulk and shear responses into additive terms whose ranges lie in $\mathbb{S}^{\text{vol}}$ and $\mathbb{S}^{\text{dev}}$ respectively. This decoupling is what allows the scalar fields $\bar{\theta}$ and $\bar{p}$ to be extracted as trace-projections of the tensor fields and promoted to independent variables with their own inf-sup condition. The deviatoric tensor fields $\bar{\boldsymbol{\epsilon}} \in H(\text{Curl}; \mathbb{S}^{\text{dev}})$ and $\bar{\boldsymbol{s}} \in H(\text{Div}; \mathbb{S}^{\text{dev}})$ inherit conformity from the trace-free projection of the full SEC spaces.

B.4 Discrete Spaces and Commuting Projections

The finite-dimensional analog of (46) is a sequence of conforming spaces $V_{\boldsymbol{\varphi},h}$, $V_{\bar{\mathbf{e}},h}$, $V_{\bar{\mathbf{s}},h}$, $V_{\bar{\mathbf{p}},h}$ on a tetrahedral mesh together with interpolation operators $\Pi_{\boldsymbol{\varphi}}, \Pi_{\bar{\mathbf{e}}}, \Pi_{\bar{\mathbf{s}}}, \Pi_{\bar{\mathbf{p}}}$ that commute with the differential operators. For sufficiently smooth fields,

$$\begin{array}{ccccccc}
 H^1(B; \mathbb{R}^3) & \xrightarrow{\text{sym Grad}} & H(\text{Curl}; \mathbb{S}) & \xrightarrow{\text{inc}} & H(\text{Div}; \mathbb{S}) & \xrightarrow{\text{Div}} & L^2(B; \mathbb{R}^3) \\
 \Pi_{\boldsymbol{\varphi}} \downarrow & & \Pi_{\bar{\mathbf{e}}} \downarrow & & \Pi_{\bar{\mathbf{s}}} \downarrow & & \Pi_{\bar{\mathbf{p}}} \downarrow \\
 V_{\boldsymbol{\varphi},h} & \xrightarrow{\text{sym Grad}} & V_{\bar{\mathbf{e}},h} & \xrightarrow{\text{inc}} & V_{\bar{\mathbf{s}},h} & \xrightarrow{\text{Div}} & V_{\bar{\mathbf{p}},h}.
 \end{array} \tag{53}$$

The commuting diagram delivers two structural consequences: the discrete complex is itself exact (modulo the kernel of $\Pi_{\boldsymbol{\varphi}}$), and the inf-sup stability of the coupled system can be read off the continuous complex by pulling stability through the projections. This is the central result of finite-element exterior calculus [5]: cochain projections that commute with the differential operators transfer stability from the continuous to the discrete level.

Two families of discrete symmetric-tensor spaces realize the commuting diagram (53) on tetrahedra.

1. **Strongly-symmetric elements.** The Arnold–Awanou–Winther family [4] enforces symmetry of the tensor field pointwise through a vertex-edge-face-cell hierarchy of degrees of freedom. The $H(\text{Div}; \mathbb{S})$ element of lowest order on a tetrahedron uses polynomial degree $k = 3$ and 162 degrees of freedom per element. The construction is geometrically clean — symmetry is a property of the finite element space — at the cost of a high local dimension at low polynomial order.
2. **Weakly-symmetric elements.** The Arnold–Falk–Winther family [3] trades pointwise symmetry for a smaller local dimension by enforcing symmetry variationally. A skew-symmetric Lagrange multiplier $\mathbf{W} \in L^2(B; \mathbb{K})$, with $\mathbb{K}$ the space of skew-symmetric second-order tensors, imposes the constraint $\int_B \mathbf{W} : \boldsymbol{\tau} \, dV = 0$ on the tensor field, and the $H(\text{Div})$ space itself need not be pointwise-symmetric. The resulting mixed element uses substantially fewer degrees of freedom at low polynomial order, at the cost of adding the skew multiplier to the system.

Both families admit cochain projections compatible with (53) and are established stable discretizations of the symmetric elasticity complex. The present formulation inherits inf-sup stability from whichever family is selected in `ReferenceFiniteElements.jl` [7]; the choice affects the element-local assembly but does not alter the variational formulation (7) or the reduced residual (17).

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